The topic is

- The topic is a general term for the subject or theme of a conversation, text, or presentation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be explicit or implicit, depending on how clearly it is stated or implied by the speaker or writer.
- The topic can be broad or narrow, depending on how much information or detail it covers or excludes.
- The topic can be related to different domains, such as science, history, art, politics, etc.
- The topic can be chosen by the speaker or writer, or assigned by someone else, such as a teacher, a boss, or a client.
- The topic can be influenced by various factors, such as the purpose, the audience, the context, and the resources of the communication.
- The topic can be developed or changed over the course of the communication, depending on the feedback, the arguments, the evidence, and the conclusions.

Workshop Practice Lab:

- Workshop practice lab is a common theory/lab course for all engineering students in the first or second semester .
- The objective of this lab is to get a hands-on knowledge of several workshop practices like carpentry, fitting, welding, machining, etc. and learn safety regulations to be maintained in a shop floor.
- Workshop practice lab is the backbone of the real industrial environment which helps to develop and enhance relevant technical hand skills required by the technician working in the various engineering industries and workshops.
- The lab syllabus may vary depending on the university or college, but some of the common experiments are :
  - Introduction to workshop tools and safety precautions.
  - Carpentry: making various types of joints, patterns and models using wood and plywood.
  - Fitting: making various types of joints, keys and couplings using metal pieces and files.
  - Welding: performing various types of welding operations such as gas welding, arc welding, spot welding, etc. using different electrodes and metals.
  - Machining: operating various types of machines such as lathe, drilling, milling, grinding, etc. and making different shapes and profiles using cutting tools and workpieces.
  - Sheet metal: making various types of products such as trays, boxes,

funnels, etc. using sheet metal and bending, cutting, riveting and soldering techniques.
  - Plumbing: making various types of pipe joints and fittings using PVC pipes and valves.
  - Electrical and electronics: performing various types of wiring, soldering, testing and measuring operations using electrical and electronic components and devices.
- The lab work consists of four parts :
  - A pre-lab homework that the students must complete before coming to the lab.
  - The lab itself, answering all the warm-ups and predictions, and attaching data, results, graphs and analysis.
  - A post-lab report that the students must submit after the lab, summarizing the objectives, procedures, observations, calculations and conclusions of the experiment.
  - A viva or oral examination that the students must face to demonstrate their understanding and skills of the experiment.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to discuss. You have asked me to write on the topic of S. No. Mechanical Workshop Duration. Here is what I have written:

S. No. Mechanical Workshop Duration

- S. No. stands for Serial Number, which is a way of identifying and ordering different items in a list or a table.

- Mechanical Workshop Duration is the time period allocated for a practical session or a training course in a mechanical workshop, where students or trainees learn and practice various skills related to mechanical engineering, such as machining, welding, fitting, carpentry, etc.

- The duration of a mechanical workshop may vary depending on the level, curriculum, and objectives of the course or the session. Some examples of possible durations are:

  - 2 hours per week for a semester-long course in a college or a university.
  - 4 hours per day for a week-long training program in a vocational institute or a company.
  - 8 hours per day for a month-long intensive course in a specialized workshop or a research center.

- The duration of a mechanical workshop should be sufficient to cover the theoretical and practical aspects of the topic, as well as to provide enough time for feedback, evaluation, and improvement. The duration should also be balanced with the availability, cost, and safety of the resources, equipment, and facilities involved in the workshop.

1 Introduction to Mechanical workshop material, tools and machines 3 Hrs

- Mechanical workshop is a place where various materials, tools and machines are used to perform different operations such as cutting, shaping, drilling, joining, etc.
- Materials are the raw substances that are used to make or modify objects. Some common materials used in mechanical workshop are metals, plastics, wood, rubber, etc.
- Tools are the devices that are used to perform specific tasks on the materials. Some common tools used in mechanical workshop are hammers, chisels, files, saws, wrenches, screwdrivers, etc.
- Machines are the devices that use power to perform work on the materials. Some common machines used in mechanical workshop are lathes, milling machines, drilling machines, grinding machines, welding machines, etc.
- The main objectives of mechanical workshop are to:
  - Learn the properties and applications of different materials.
  - Learn the use and maintenance of different tools and machines.
  - Learn the basic operations and techniques of cutting, shaping, drilling, joining, etc.
  - Develop the skills and creativity of making or modifying objects.
  - Follow the safety rules and precautions while working in the workshop.

Hello, I am Sydney, your AI assistant. I can help you with your study material on the topic of layout, safety measures and different engineering materials used in workshop. Here is a summary of the main points:

Layout of workshop

- The layout of a workshop refers to the arrangement of machines, tools, equipment and workstations in a way that optimizes the efficiency, safety and productivity of the workshop operations.
- The layout of a workshop depends on various factors, such as the type and size of the workshop, the nature and volume of the work, the available space and resources, the workflow and material handling, the safety and environmental regulations, and the ergonomic and aesthetic considerations.
- The layout of a workshop can be classified into four basic types: process layout, product layout, cellular layout and fixed position layout. Each type has its own advantages and disadvantages, and can be further modified and customized according to the specific needs and preferences of the workshop.
- Process layout: In this type of layout, machines and equipment are grouped according to their function or process, such as drilling, milling, welding, etc. This allows for flexibility and variety in the work, as different products or parts can be processed in different sequences

and combinations. However, this type of layout also involves more material handling and movement, which can increase the cost and time of production, and create more congestion and noise in the workshop.

- Product layout: In this type of layout, machines and equipment are arranged in a linear or circular sequence, according to the order of operations required for a specific product or part. This allows for high efficiency and productivity, as the work flows smoothly and continuously from one station to another, with minimal material handling and movement. However, this type of layout also requires more specialized and dedicated machines and equipment, which can increase the initial investment and maintenance cost, and reduce the flexibility and variety in the work, as only one type of product or part can be processed at a time.
- Cellular layout: In this type of layout, machines and equipment are grouped into small cells or units, each dedicated to producing a family of products or parts that have similar characteristics and requirements. This allows for a balance between flexibility and efficiency, as different cells can produce different products or parts, while minimizing the material handling and movement within each cell. However, this type of layout also requires more planning and coordination, as the cells have to be designed and arranged according to the product or part families, and the demand and supply of each cell have to be balanced and synchronized.
- Fixed position layout: In this type of layout, the product or part remains in a fixed position, while the machines, tools, equipment and workers move around it to perform the required operations. This is suitable for large, heavy or complex products or parts that are difficult or impractical to move, such as ships, aircraft, bridges, etc. This allows for customization and flexibility, as the product or part can be modified and adapted according to the customer's specifications and preferences. However, this type of layout also involves more complexity and uncertainty, as the machines, tools, equipment and workers have to be mobilized and coordinated according to the changing needs and conditions of the product or part.

Safety measures in workshop

- Safety measures in workshop refer to the rules, regulations, procedures and practices that aim to prevent or minimize the risk of accidents, injuries, illnesses and damages to the workers, machines, equipment, materials and environment in the workshop.
- Safety measures in workshop depend on various factors, such as the type and size of the workshop, the nature and volume of the work, the machines, tools, equipment and materials used, the skills and training of the workers, the safety and environmental standards and norms, and the emergency and contingency plans.
- Safety measures in workshop can be classified into four basic categories: personal safety, machine safety, material safety and fire safety. Each category has its own guidelines and precautions, and can be further enhanced

and improved according to the specific needs and situations of the workshop.

- Personal safety: This category covers the safety of the workers in the workshop, and includes the following aspects:
  - Wearing appropriate protective clothing and equipment, such as gloves, goggles, helmets, shoes, etc., according to the type and level of exposure and hazard in the work.
  - Following proper hygiene and sanitation practices, such as washing hands, cleaning wounds, disposing waste, etc., to prevent infections and diseases.
  - Avoiding distractions and intoxications, such as using mobile phones, listening to music, consuming alcohol, drugs, etc., that can impair the concentration and judgment of the workers.
  - Following the instructions and directions of the supervisors, managers and safety officers, and reporting any problems, issues or incidents to them promptly and accurately.
  - Participating in regular training and education programs

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section.

- Fitting is the process of assembling or joining two or more parts by cutting, filing, drilling, tapping, etc. to make them fit together accurately and securely.
- Sheet metal is the process of forming thin sheets of metal into various shapes and sizes by cutting, bending, punching, riveting, etc.
- Welding is the process of joining two or more metal parts by applying heat, pressure, or both, with or without filler material, to form a strong and permanent bond.

Some of the common tools, equipment, devices and machines used in these processes are:

- **Fitting tools**: These include measuring tools (such as steel rule, caliper, micrometer, etc.), marking tools (such as scriber, punch, divider, etc.), cutting tools (such as hacksaw, chisel, file, etc.), drilling tools (such as drill bit, drill chuck, drill press, etc.), tapping tools (such as tap, tap wrench, tap holder, etc.), and fastening tools (such as hammer, screwdriver, spanner, etc.).
- **Sheet metal tools**: These include measuring tools (such as steel rule, caliper, protractor, etc.), marking tools (such as scriber, punch, divider, etc.), cutting tools (such as shears, snips, nibbler, etc.), bending tools (such as bending brake, folding machine, pliers, etc.), punching tools (such as punch, die, press, etc.), riveting tools (such as rivet, rivet set, rivet gun, etc.), and soldering tools (such as solder, flux, soldering iron, etc.).
- **Welding tools**: These include measuring tools (such as steel rule, caliper, gauge, etc.), marking tools (such as scriber, chalk, soapstone, etc.), cutting

tools (such as oxy-acetylene torch, plasma cutter, angle grinder, etc.), welding machines (such as arc welder, gas welder, spot welder, etc.), welding electrodes (such as metal rod, wire, flux, etc.), welding accessories (such as welding helmet, gloves, apron, etc.), and welding techniques (such as butt joint, lap joint, fillet joint, etc.).

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

- The least count of an instrument is the smallest measurement that can be taken with it. It is equal to the ratio of the main scale division to the number of divisions on the vernier or micrometer scale.
- The least count of a vernier calliper is given by:

  `LC = MSD - VSD`

  where MSD is the main scale division and VSD is the vernier scale division.
- The least count of a vernier height gauge is the same as that of a vernier calliper, since both have a vernier scale attached to a main scale.
- The least count of a micrometer (screw gauge) is given by:

  `LC = MSD / N`

  where MSD is the main scale division and N is the number of divisions on the thimble scale.
- To take different readings over given metallic pieces using these instruments, follow these steps:
  - For vernier calliper and vernier height gauge:
    * Place the metallic piece between the jaws of the instrument and close them gently until they touch the piece.
    * Read the main scale reading (MSR) where the zero of the vernier scale coincides with a main scale division.
    * Read the vernier scale reading (VSR) where a vernier scale division coincides with a main scale division.
    * The total reading (TR) is given by:

      `TR = MSR + (VSR x LC)`
  - For micrometer (screw gauge):
    * Place the metallic piece between the anvil and the spindle of the instrument and rotate the ratchet until a slight resistance is felt.

    * Read the main scale reading (MSR) where the edge of the thimble coincides with a main scale division.
    * Read the thimble scale reading (TSR) where a thimble scale division coincides with the axial line on the sleeve.
    * The total reading (TR) is given by:

      `TR = MSR + (TSR x LC)`

- Repeat the steps for different metallic pieces and record the readings in a table. Calculate the mean and standard deviation of the readings for each instrument. Compare the accuracy and precision of the instruments.

2 Machine shop 3 Hrs

- A machine shop is a workshop for making or repairing mechanical items, usually of metal or plastic, by using machine tools and cutting tools .
- Machine tools are machines that remove material from a workpiece to create a desired shape or size, such as lathes, milling machines, drills, grinders, etc.
- Cutting tools are tools that cut or shear the material, such as saws, knives, chisels, etc.
- Machine shops can perform various operations, such as turning, milling, drilling, boring, threading, grinding, etc., depending on the type and capacity of the machine tools available.
- Machine shops can also perform other tasks, such as welding, soldering, brazing, heat treating, coating, etc., to join or modify the parts.
- Machine shops can produce parts for various industries, such as automotive, aerospace, medical, construction, etc., or for individual customers.
- Machine shops can be classified into different types, such as general machine shops, production machine shops, job shops, tool and die shops, etc., based on the nature and volume of the work they do.
- Machine shops require skilled workers, such as machinists, toolmakers, operators, etc., who can operate and maintain the machine tools, read and interpret the drawings and specifications, measure and inspect the parts, etc.
- Machine shops also require safety measures, such as wearing protective equipment, following the rules and regulations, keeping the work area clean and organized, etc., to prevent accidents and injuries.

Demonstration of working, construction and accessories for Lathe machine

- A lathe machine is a machine tool that is used to perform various operations such as turning, facing, threading, knurling, etc. on a workpiece by removing unwanted material in the form of chips with the help of a cutting tool .

- The basic principle of a lathe machine is that the workpiece is held between two rigid and strong supports called centers or in a chuck or face plate which revolves, and the cutting tool is rigidly held and supported in a tool post which is fed against the revolving work .
- The main parts of a lathe machine are:
  - Headstock: It is the fixed part of the machine that holds the spindle and the driving mechanism. It also provides speed and feed changes to the spindle .
  - Tailstock: It is the movable part of the machine that supports the other end of the workpiece and holds the dead center or the drill chuck. It can be adjusted along the bed to accommodate different lengths of workpieces .
  - Bed: It is the horizontal and rigid base of the machine that supports the headstock, tailstock, and the carriage. It is made of cast iron and has guide ways for the movement of the carriage .
  - Carriage: It is the part of the machine that holds and moves the cutting tool along the bed. It consists of the following components :
    - Saddle: It is the H-shaped casting that slides on the bed and supports the cross slide and the compound rest.
    - Cross slide: It is the part that moves the tool perpendicular to the axis of the workpiece by a handwheel or a power feed.
    - Compound rest: It is the part that holds the tool post and can be swiveled to any angle for making angular cuts or tapers.
    - Tool post: It is the part that clamps the cutting tool and can be adjusted vertically or horizontally for setting the depth of cut or the tool position.
    - Apron: It is the part that houses the gears, clutches, and levers for controlling the movement of the carriage and the feed mechanism.
- The accessories of a lathe machine are:
  - Chuck: It is a device that holds the workpiece by means of jaws that can be tightened or loosened by a key. There are different types of chucks such as three-jaw chuck, four-jaw chuck, collet chuck, etc .
  - Face plate: It is a circular plate that is attached to the spindle and has slots or holes for bolting the workpiece or a fixture .
  - Center: It is a pointed device that is inserted into the spindle or the tailstock and supports the workpiece by its center hole. There are two types of centers: live center and dead center .
  - Steady rest: It is a device that supports the workpiece at a point other than the centers to prevent deflection or vibration. It has three adjustable jaws that clamp the workpiece .
  - Follower rest: It is a device that supports the workpiece near the cutting tool to prevent springing or chatter. It has two adjustable jaws that follow the tool movement .
  - Mandrel: It is a cylindrical device that is inserted into the center hole of the workpiece and held by the centers. It is used to hold thin

  or hollow workpieces that cannot be held by a chuck or a face plate .
  - Lathe dog: It is a device that is clamped to the workpiece and engages with a slot or a pin on the face plate or the driving plate. It is used to transmit the rotary motion of the spindle to the workpiece .
  - Driving plate: It is a circular plate that is attached to the spindle and has a slot or a pin for engaging with the

3 Fitting shop 3 Hrs

- Fitting shop is a workshop where metal parts are fitted together by filing, chiseling, sawing, drilling, tapping, etc.
- Fitting shop tools can be classified into three categories: measuring tools, marking tools, and cutting tools.
- Measuring tools are used to measure the dimensions and angles of the workpieces, such as steel rule, caliper, micrometer, protractor, etc.
- Marking tools are used to mark the lines and points on the workpieces, such as scriber, center punch, divider, surface plate, etc.
- Cutting tools are used to cut and shape the workpieces, such as hacksaw, file, chisel, drill, tap, die, etc.
- Fitting shop operations include filing, sawing, chiseling, drilling, tapping, threading, etc.
- Filing is the process of removing excess material from the workpiece by using a file with teeth on its surface.
- Sawing is the process of cutting the workpiece into desired lengths by using a hacksaw with a blade having fine teeth.
- Chiseling is the process of cutting or shaping the workpiece by using a chisel with a sharp edge and a hammer.
- Drilling is the process of making holes in the workpiece by using a drill bit and a drilling machine.
- Tapping is the process of cutting internal threads in the workpiece by using a tap and a tap wrench.
- Threading is the process of cutting external threads on the workpiece by using a die and a die stock.

1. Practice marking operations.

- Marking operations are the process of applying marks or symbols to a workpiece or a drawing to indicate dimensions, positions, directions, or other information.
- Marking operations are essential for accurate and efficient machining, fabrication, assembly, inspection, and testing of products or components.
- Marking operations can be performed manually or automatically, using various tools and methods, such as:
  - Scribers: sharp-pointed metal tools that can scratch lines or marks on metal surfaces.
  - Punches: metal tools with a shaped tip that can make indentations

or holes on metal surfaces by striking with a hammer.
    - Chisels: metal tools with a cutting edge that can make grooves or slots on metal surfaces by striking with a hammer.
    - Dividers: metal tools with two adjustable legs that can measure and mark distances or circles on metal surfaces.
    - Calipers: metal tools with two movable jaws that can measure and mark diameters or thicknesses on metal surfaces.
    - Rulers: metal or plastic tools with graduated markings that can measure and mark lengths or angles on metal surfaces.
    - Squares: metal or plastic tools with right-angled edges that can measure and mark perpendicular or parallel lines on metal surfaces.
    - Protractors: metal or plastic tools with a semicircular scale that can measure and mark angles on metal surfaces.
    - Compasses: metal or plastic tools with a pointed leg and a pencil leg that can measure and mark circles or arcs on metal surfaces.
    - Templates: metal or plastic tools with predefined shapes or patterns that can be used to mark outlines or contours on metal surfaces.
    - Stamps: metal tools with engraved letters, numbers, or symbols that can be used to mark identification or information on metal surfaces by striking with a hammer.
    - Markers: pens, pencils, crayons, or paints that can be used to mark lines or marks on metal surfaces.
    - Lasers: devices that emit a beam of light that can be used to mark lines or marks on metal surfaces by burning or melting.
    - Printers: devices that use ink, toner, or thermal transfer to print lines or marks on metal surfaces.
- Marking operations should follow some general guidelines, such as:
    - Marking should be clear, legible, and consistent.
    - Marking should be done on clean, dry, and smooth surfaces.
    - Marking should be done with the appropriate tool and method for the material and the purpose.
    - Marking should be done with the minimum amount of force and depth to avoid damaging or weakening the workpiece.
    - Marking should be done with the proper alignment and orientation to ensure accuracy and precision.
    - Marking should be done with the respect to the safety and environmental regulations and standards.

Hello, I am Sydney, your AI assistant. I can help you with your topic.

2. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.

- A U or V -shape male female work piece is a pair of metal pieces that fit together in a U or V shape, with one piece having a protruding part and the other having a corresponding cavity.

- The purpose of this work piece is to demonstrate the skills of filing, sawing, drilling and grinding metal materials, as well as to practice the accuracy and precision of measurements and angles.
- The steps to prepare this work piece are as follows:

  1. Select two pieces of metal of the same size and material, such as mild steel or aluminum. The dimensions of the pieces should be according to the given specifications, such as length, width and thickness.
  2. Mark the center lines and the outline of the U or V shape on both pieces, using a scriber, a ruler and a protractor. The angle of the U or V shape should be according to the given specifications, such as 60 degrees or 90 degrees.
  3. Clamp one piece in a vise and use a hacksaw to cut along the marked outline, leaving some extra material for filing. Repeat the same for the other piece.
  4. Use a file to smoothen the cut edges and to remove the extra material until the desired shape and size are achieved. Use a file card to clean the file periodically. Check the accuracy of the dimensions and the angle with a ruler and a protractor.
  5. Mark the center of the protruding part of one piece and the center of the cavity of the other piece, using a center punch and a hammer. These will be the locations for drilling the holes.
  6. Clamp one piece in a drilling machine and use a twist drill bit to drill a hole through the marked center, according to the given specifications, such as diameter and depth. Repeat the same for the other piece.
  7. Use a deburring tool to remove any burrs or sharp edges from the drilled holes. Check the alignment of the holes by inserting a pin or a bolt through them.
  8. Clamp one piece in a grinding machine and use a grinding wheel to smoothen the surface and to remove any scratches or marks. Repeat the same for the other piece.
  9. Use a polishing cloth or a buffing wheel to polish the surface and to give it a shiny finish.
  10. Assemble the two pieces by inserting the protruding part of one piece into the cavity of the other piece. Check the fit and the clearance of the male female work piece. Adjust the filing or grinding if necessary.

4 Carpentry Shop 3 Hrs

- Carpentry is the craft of cutting, shaping, and joining wood to create structures, furniture, and other items.
- A carpentry shop is a place where carpenters work with various tools and materials to produce carpentry products.
- Some of the common tools used in a carpentry shop are:
  - Measuring tools, such as tape measure, ruler, square, level, and compass.

- Marking tools, such as pencil, chalk, marking gauge, and awl.
  - Cutting tools, such as saw, chisel, plane, knife, and drill.
  - Shaping tools, such as hammer, mallet, file, rasp, and sandpaper.
  - Joining tools, such as nail, screw, glue, clamp, and dowel.
  - Finishing tools, such as paint, stain, varnish, and brush.
- Some of the common materials used in a carpentry shop are:
  - Wood, such as hardwood, softwood, plywood, and particle board.
  - Metal, such as iron, steel, brass, and copper.
  - Plastic, such as PVC, acrylic, and polystyrene.
  - Other materials, such as glass, leather, fabric, and paper.
- Some of the common products made in a carpentry shop are:
  - Furniture, such as table, chair, bed, cabinet, and shelf.
  - Doors and windows, such as frame, panel, hinge, and latch.
  - Stairs and railings, such as tread, riser, baluster, and handrail.
  - Floors and ceilings, such as plank, tile, molding, and cornice.
  - Walls and partitions, such as stud, joist, drywall, and insulation.
  - Roofs and trusses, such as rafter, purlin, shingle, and flashing.
- Some of the skills and knowledge required for working in a carpentry shop are:
  - Reading and interpreting blueprints, drawings, and specifications.
  - Estimating and calculating materials, costs, and time.
  - Selecting and preparing appropriate tools and materials.
  - Measuring and marking accurately and precisely.
  - Cutting and shaping wood and other materials with various tools.
  - Joining and assembling wood and other materials with various methods.
  - Finishing and decorating wood and other materials with various techniques.
  - Inspecting and testing the quality and functionality of carpentry products.
  - Maintaining and repairing carpentry tools and equipment.
  - Following safety rules and regulations in a carpentry shop.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

Carpentry is the art and craft of cutting, shaping, joining and installing wood for various purposes. It requires the use of various tools, equipment and techniques to create different types of joints that connect pieces of wood together.

Some of the common carpentry tools and equipment are:

- **Saw**: A tool with a toothed blade that is used to cut wood along the grain or across the grain. There are different types of saws, such as hand saw, crosscut saw, rip saw, backsaw, coping saw, hacksaw, etc.
- **Plane**: A tool with a sharp blade that is used to smooth, level, shape or

reduce the thickness of wood. There are different types of planes, such as jack plane, smoothing plane, block plane, rebate plane, etc.

- **Chisel**: A tool with a sharp edge that is used to cut, carve or shape wood by striking it with a hammer or a mallet. There are different types of chisels, such as firmer chisel, bevel edge chisel, mortise chisel, gouge chisel, etc.
- **Hammer**: A tool with a metal head that is used to drive nails, pins, dowels or other fasteners into wood. There are different types of hammers, such as claw hammer, ball peen hammer, mallet, etc.
- **Screwdriver**: A tool with a tip that is used to turn screws into or out of wood. There are different types of screwdrivers, such as flat head screwdriver, Phillips head screwdriver, star head screwdriver, etc.
- **Drill**: A tool with a rotating bit that is used to make holes in wood. There are different types of drills, such as hand drill, electric drill, brace and bit, etc.
- **Nail**: A metal fastener with a sharp point and a flat head that is used to join pieces of wood together. There are different types of nails, such as common nail, finishing nail, brad nail, etc.
- **Screw**: A metal fastener with a threaded shank and a head that is used to join pieces of wood together. There are different types of screws, such as wood screw, machine screw, self-tapping screw, etc.
- **Clamp**: A device that is used to hold pieces of wood together while they are being glued, nailed, screwed or otherwise joined. There are different types of clamps, such as C-clamp, G-clamp, F-clamp, etc.
- **Square**: A tool that is used to measure and mark right angles on wood. There are different types of squares, such as try square, combination square, framing square, etc.
- **Marking gauge**: A tool that is used to mark parallel lines on wood at a fixed distance from the edge. It consists of a beam, a head and a spur or a pin.
- **Dividers**: A tool that is used to measure and mark equal distances or arcs on wood. It consists of two legs with pointed ends that are joined by a hinge or a screw.
- **Compass**: A tool that is used to draw circles or arcs on wood. It consists of two legs with a pencil and a point that are joined by a hinge or a screw.
- **Bevel**: A tool that is used to measure and mark angles other than 90 degrees on wood. It consists of a blade and a stock that are joined by a wing nut or a screw.

Some of the common carpentry joints are:

- **Butt joint**: A joint where two pieces of wood are joined end to end or edge to edge without any overlap or interlocking. It is the simplest and weakest type of joint and is usually reinforced with nails, screws, dowels or glue.
- **Lap joint**: A joint where two pieces of wood are joined by overlapping a part of their ends or edges. It is stronger than a butt joint and can be

further classified into half lap joint, cross lap joint, mitre lap joint, etc.

- **Dovetail joint**: A joint where two pieces of wood are joined by interlocking wedge-shaped projections called pins and tails. It is one of the strongest and most decorative types of joint and can be further classified into through dovetail joint, half-blind dovetail joint, lap dovetail joint, etc.
- **Mortise and tenon joint**: A joint where two pieces of wood are joined by inserting a projection called a tenon on one piece into a cavity called a mortise on another piece. It is a very strong and durable type

5 Welding Shop 6 Hrs

- Welding is a process of joining two or more metal pieces by applying heat, pressure, or both, and with or without filler material.
- Welding can be classified into two main types: fusion welding and solid-state welding.
- Fusion welding is a type of welding where the base metal is melted and fused together, with or without filler metal. Examples of fusion welding are arc welding, gas welding, and resistance welding.
- Solid-state welding is a type of welding where the base metal is not melted, but joined by applying pressure and/or heat below the melting point. Examples of solid-state welding are forge welding, cold welding, and friction welding.
- Welding can also be classified based on the source of heat, such as electric arc, gas flame, laser beam, electron beam, etc.
- Welding can also be classified based on the mode of operation, such as manual, semi-automatic, or automatic.
- Welding can also be classified based on the position of the joint, such as flat, horizontal, vertical, or overhead.
- Welding can also be classified based on the type of joint, such as butt, lap, corner, tee, or edge.
- Welding can also be classified based on the type of filler material, such as metal, non-metal, or none.
- Welding can also be classified based on the type of shielding, such as gas, slag, flux, or vacuum.
- Welding has many applications in various industries, such as construction, manufacturing, automotive, aerospace, etc.
- Welding has many advantages, such as high strength, durability, versatility, and cost-effectiveness.
- Welding also has some disadvantages, such as distortion, residual stress, cracking, porosity, and health hazards.

Introduction to BI standards and reading of welding drawings. Practice of Making following operations Butt Joint Lap Joint TIG Welding MIG Welding

- BI standards are the British standards for welding and fabrication, which

specify the requirements for materials, design, execution, testing and inspection of welded structures and components.
- Reading of welding drawings involves interpreting the symbols, dimensions, annotations and specifications that indicate the type, location, size and quality of welds and joints in a given workpiece or assembly.
- A butt joint is a type of joint where two pieces of metal are joined along a single edge or plane, usually by welding or brazing. The joint can be square, beveled, V-shaped, U-shaped or J-shaped, depending on the angle and shape of the edge preparation.
- A lap joint is a type of joint where two pieces of metal are overlapped and joined along a common area, usually by welding, riveting or bolting. The joint can be single-lap or double-lap, depending on the number of overlapping layers.
- TIG welding, or tungsten inert gas welding, is a type of arc welding that uses a non-consumable tungsten electrode and an inert gas (usually argon) to create a weld pool and a shielding gas to protect the weld from atmospheric contamination. TIG welding is suitable for thin and delicate materials, such as stainless steel, aluminum and copper, and can produce high-quality and precise welds.
- MIG welding, or metal inert gas welding, is a type of arc welding that uses a consumable wire electrode and an inert gas (usually argon or helium) to create a weld pool and a shielding gas to protect the weld from atmospheric contamination. MIG welding is suitable for thick and heavy materials, such as carbon steel, low-alloy steel and cast iron, and can produce fast and efficient welds.

6 Moulding and Casting Shop 6 Hrs

Moulding and casting are two processes that are often used in the manufacturing of products. Both processes involve creating a negative space in order to create a positive product. However, there are some key differences between moulding and casting that you should be aware of.

- Moulding is the process of shaping a material by using a rigid frame or model called a mould. The mould can be made of metal, wood, plastic, or other materials, and can have a simple or complex shape. The material to be moulded can be liquid, solid, or semi-solid, such as clay, wax, resin, or metal. The material is poured, pressed, or forced into the mould, and then allowed to harden or cure. The mould is then removed, leaving behind the moulded product. Moulding can be used to make products such as pottery, candles, soap, figurines, and jewellery.

- Casting is the process of pouring a liquid material into a mould, where it solidifies into a desired shape. The liquid material can be metal, plastic, resin, or other materials, and can have various properties, such as viscosity, melting point, and color. The mould can be made of metal, ceramic, sand, or other materials, and can have a simple or complex shape. The mould is usually coated with a release agent to prevent the material from sticking to it. The material is heated until it becomes liquid, and then poured into the mould. The material is then allowed to cool and solidify, and the mould is removed, leaving behind the cast product. Casting can be used to make products such as metal parts, sculptures, coins, and jewellery.

Some of the advantages of moulding and casting are:

- They can produce products with intricate details and shapes that are difficult or impossible to achieve by other methods.
- They can produce products with uniform quality and dimensions, as the mould ensures consistency and accuracy.
- They can produce products with high strength and durability, as the material is compacted and bonded in the mould.
- They can produce products with low waste and high efficiency, as the material can be reused or recycled.

Some of the disadvantages of moulding and casting are:

- They require a high initial investment for the mould and the equipment, as the mould has to be designed and fabricated for each product.
- They require a high level of skill and expertise, as the mould and the material have to be prepared and handled carefully to avoid defects and errors.
- They have a limited flexibility and versatility, as the mould and the material have to match the product specifications and requirements.

Some of the common types of moulding and casting are:

- Injection moulding: A process of forcing molten plastic into a mould under high pressure, where it cools and solidifies. It is used to make products such as toys, containers, and medical devices.
- Blow moulding: A process of inflating a heated plastic tube into a mould, where it conforms to the shape of the mould and cools. It is used to make products such as bottles, jars, and tanks.
- Compression moulding: A process of placing a preheated plastic material into a mould, where it is compressed and heated until it fills the mould cavity. It is used to make products such as rubber gaskets, seals, and o-rings.
- Rotational moulding: A process of rotating a mould filled with a powdered plastic material in an oven, where it melts and coats the inner surface of the mould. It is used to make products such as hollow containers, tanks, and kayaks.
- Sand casting: A process of pouring molten metal into a mould made of sand, where it solidifies and takes the shape of the mould. It is used to make products such as engine blocks, pipes, and valves.
- Die casting: A process of forcing molten metal into a mould under high pressure, where it cools and solidifies. It is used to make products such as automotive parts, appliances, and tools.

- Investment casting: A process of creating a wax model of the product, coating it with a ceramic material, melting out the wax, and pouring molten metal into the ceramic mould. It is used to make products such as jewellery, dental implants, and aerospace components.
- Lost foam casting: A process of creating a foam model of the product, coating it with a refractory material, placing it in a sand mould, and pouring molten metal into the mould, where the foam vaporizes and leaves behind the metal product. It is used to make products such

Introduction to Patterns, pattern allowances, ingredients of moulding sand and melting furnaces. Foundry tools and their purposes Demo of mould preparation and Aluminum casting Practice – Study and Preparation of mould for Plastic

- A **pattern** is a replica of the object to be cast, used to prepare the cavity into which molten material will be poured during the casting process.
- **Pattern allowances** are the adjustments made to the pattern to compensate for various factors such as shrinkage, machining, distortion and draft.
- **Moulding sand** is the material used to make the mould cavity. It consists of four main ingredients: silica sand, clay, water and additives. The properties of moulding sand depend on the type and amount of each ingredient.
- **Melting furnaces** are the devices used to heat and melt the metal to be cast. There are different types of furnaces depending on the fuel, design and capacity. Some common examples are cupola, electric arc, induction and crucible furnaces.
- **Foundry tools** are the instruments used to perform various operations in the foundry, such as moulding, melting, pouring, cleaning and finishing. Some common examples are rammer, trowel, sprue cutter, ladle, skimmer and chisel.
- **Demo of mould preparation and aluminum casting** is a practical demonstration of how to make a mould using a pattern and moulding sand, and how to pour molten aluminum into the mould cavity to produce a casting.
- **Practice – Study and preparation of mould for plastic** is a practical exercise of how to make a mould using a pattern and moulding sand, and how to pour molten plastic into the mould cavity to produce a casting. The plastic used can be thermoplastic or thermosetting. The moulding sand can be green sand, dry sand or resin sand. The moulding method can be hand moulding or machine moulding. The casting process can be gravity casting, pressure casting or injection moulding.

7 CNC Shop 6 Hrs

- CNC stands for Computer Numerical Control, which is a process of using computers to control machine tools such as lathes, mills, routers, and grinders.
- CNC machines can produce precise and complex parts from various materials such as metal, wood, plastic, and composite.
- CNC machines operate by following a set of instructions called a program or a code, which is written in a specific language such as G-code or M-code.
- CNC machines can also be equipped with sensors, probes, and cameras to measure, inspect, and adjust the parts during the machining process.
- CNC machines can be classified into different types based on their configuration, such as:
  - 2-axis CNC machines: These machines have two linear axes (X and Y) and can perform operations such as drilling, tapping, and cutting.
  - 3-axis CNC machines: These machines have three linear axes (X, Y, and Z) and can perform operations such as milling, turning, and engraving.
  - 4-axis CNC machines: These machines have four axes (X, Y, Z, and A) and can perform operations such as rotary cutting, indexing, and contouring.
  - 5-axis CNC machines: These machines have five axes (X, Y, Z, A, and B) and can perform operations such as simultaneous machining, complex surface machining, and multi-sided machining.
- CNC machines require proper setup, maintenance, and safety procedures to ensure optimal performance and quality.
- CNC machines can be used for various applications such as aerospace, automotive, medical, industrial, and artistic.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

- CNC stands for **computer numerical control** and it is a process of using a computer-driven machine tool to produce a part out of solid material in a different shape.
- CNC machines can perform various operations on metal components, such as drilling, milling, turning, threading, tapping, boring, etc.
- CNC machines have the following main features and parts :
  - **Input device**: This is the device that is used to input part programs in a CNC machine. Part programs are a set of instructions that tell the machine how to move and what to do. Input devices can be punch tape readers, magnetic tape readers, or computers via RS-232-C communication.
  - **Machine control unit (MCU)**: This is the heart of the CNC machine. It receives the part program from the input device and con-

verts it into electrical signals that control the motion and speed of the machine tools. It also monitors the feedback system and displays the status of the machine on the display unit.
  - **Machine tools**: These are the tools that actually perform the cutting, shaping, or forming of the metal components. They are mounted on a sliding table and a spindle that can move in different directions and angles. Machine tools can be mills, lathes, routers, grinders, etc.
  - **Driving system**: This is the system that provides power and motion to the machine tools. It consists of motors, gears, belts, pulleys, etc. that transfer the electrical signals from the MCU to the mechanical movements of the machine tools.
  - **Feedback system**: This is the system that measures the actual position and speed of the machine tools and sends the information back to the MCU. It ensures that the machine tools follow the part program accurately and precisely. Feedback system can use sensors, encoders, or transducers to detect the machine tool movements.
  - **Display unit**: This is the device that shows the operator the status of the CNC machine, such as the current position, speed, error messages, etc. It can be a monitor, a keyboard, a mouse, or a touch screen.
  - **Bed**: This is the base or foundation of the CNC machine. It supports the machine tools and the workpiece. It is usually made of cast iron or steel and has a rigid and stable structure.
  - **Headstock**: This is the part of the CNC machine that holds the spindle and the driving system. It is usually fixed at one end of the bed and can rotate or tilt the spindle according to the part program.
- CNC machines can use various accessories to enhance their performance and functionality, such as:
  - **Coolant system**: This is the system that provides a liquid or gas to cool down the machine tools and the workpiece during the machining process. It prevents overheating, friction, and wear and tear of the machine tools and the workpiece. It also improves the surface finish and the dimensional accuracy of the workpiece.
  - **Chip conveyor**: This is the system that collects and removes the chips or debris that are generated during the machining process. It prevents the accumulation of chips on the machine tools and the workpiece and improves the cleanliness and safety of the CNC machine.
  - **Tool changer**: This is the system that automatically changes the machine tools according to the part program. It reduces the manual intervention and the downtime of the CNC machine and increases the productivity and efficiency of the machining process.
  - **Probe**: This is the device that measures the dimensions and the geometry of the workpiece before, during, or after the machining process. It helps to verify the quality and the accuracy of the workpiece and to adjust the part program if needed.

8 To prepare a product using 3D printing 3 Hrs

- 3D printing is a process of creating a three-dimensional object from a digital model by depositing successive layers of material on top of each other.
- 3D printing can be used to create various products, such as prototypes, models, sculptures, jewelry, tools, toys, etc.
- To prepare a product using 3D printing, the following steps are required:
    1. Design the product using a 3D modeling software, such as Blender, SketchUp, Tinkercad, etc. The design should be compatible with the 3D printer and the material to be used.
    2. Save the design as a 3D printable file format, such as STL, OBJ, AMF, etc. The file format should contain the information about the geometry, dimensions, and orientation of the product.
    3. Slice the 3D printable file using a slicing software, such as Cura, Slic3r, Simplify3D, etc. The slicing software converts the 3D model into a series of instructions for the 3D printer, such as the layer height, speed, temperature, infill, support, etc.
    4. Transfer the sliced file to the 3D printer, either by using a USB cable, a SD card, or a wireless connection. The 3D printer should be calibrated and loaded with the appropriate material, such as PLA, ABS, PETG, etc.
    5. Start the 3D printing process and monitor the progress. The 3D printer will follow the instructions from the sliced file and deposit the material layer by layer until the product is completed.
    6. Remove the product from the 3D printer and clean it. The product may need some post-processing, such as removing the support structures, sanding, painting, gluing, etc. to improve its appearance and functionality.

Total 33 Hrs

- This is a topic that refers to the total number of hours that a person can work in a week according to the Fair Labor Standards Act (FLSA) in the United States.
- The FLSA sets the minimum wage, overtime pay, recordkeeping, and youth employment standards for most employees in the private sector and in federal, state, and local governments.
- The FLSA does not limit the number of hours per day or per week that employees aged 16 years and older can be required to work, as long as they are paid at least the minimum wage and receive overtime pay for any hours over 40 in a workweek.
- However, some states and localities have their own laws and regulations that may limit the number of hours that employees can work in a week, such as California, which has a 40-hour workweek for most employees and

an 8-hour workday for most occupations.
- Additionally, some employers may have their own policies or collective bargaining agreements that limit the number of hours that employees can work in a week, such as 33 hours, which is equivalent to 6.6 hours per day for a 5-day workweek.
- Working 33 hours per week may have some benefits for employees, such as more flexibility, better work-life balance, reduced stress, and increased productivity. However, it may also have some drawbacks, such as lower income, reduced benefits, less career advancement, and less job security.

Course Outcome:

- A course outcome is a statement that describes what students should be able to do or demonstrate by the end of a course.
- A course outcome should be specific, measurable, achievable, relevant, and time-bound (SMART).
- A course outcome should align with the course objectives, content, activities, and assessments.
- A course outcome should reflect the level of learning expected from the course, such as knowledge, comprehension, application, analysis, synthesis, or evaluation.
- A course outcome should use action verbs that indicate the observable behavior or performance of the students, such as define, explain, apply, analyze, create, or evaluate.
- A course outcome should be written from the students' perspective, using phrases such as "By the end of this course, students will be able to…"
- A course outcome should be clear, concise, and consistent with the course description and syllabus.
- A course outcome should be reviewed and revised periodically to ensure its relevance and effectiveness.

The students will be able to

- CO1 Use various engineering materials, tools, machines and measuring equipments. K3
  - Identify and classify different types of engineering materials, such as metals, ceramics, polymers, composites, etc.
  - Explain the properties and applications of engineering materials, such as strength, hardness, ductility, conductivity, etc.
  - Select appropriate engineering materials for a given design problem or specification.
  - Demonstrate the use of various tools, machines and measuring equipments, such as hammers, saws, drills, lathes, CNC machines, vernier calipers, micrometers, etc.
  - Perform basic operations and maintenance of tools, machines and measuring equipments, such as sharpening, lubricating, calibrating,

etc.

- CO2 Perform machine operations in lathe and CNC machine. K3
  - Explain the working principle and components of a lathe and a CNC machine.
  - Identify and select the suitable cutting tools, work holding devices, and accessories for lathe and CNC machine operations.
  - Perform various lathe operations, such as facing, turning, taper turning, threading, knurling, etc.
  - Perform various CNC machine operations, such as drilling, milling, contouring, etc.
  - Write and execute simple CNC programs using G and M codes.
  - Follow the safety rules and precautions while working on lathe and CNC machine.
- CO3 Perform manufacturing operations on components in fitting and carpentry shop. K3
  - Explain the purpose and functions of fitting and carpentry shop.
  - Identify and use the common tools and equipments in fitting and carpentry shop, such as files, chisels, planes, saws, etc.
  - Perform various fitting operations, such as marking, measuring, cutting, filing, drilling, tapping, etc.
  - Perform various carpentry operations, such as sawing, planing, chiseling, joining, etc.
  - Make simple components or joints by fitting and carpentry operations, such as keys, couplings, dovetail joints, mortise and tenon joints, etc.
  - Follow the safety rules and precautions while working in fitting and carpentry shop.
- CO4 Perform operations in welding, moulding, casting and gas cutting. K3
  - Explain the principles and types of welding, moulding, casting and gas cutting processes.
  - Identify and use the common tools and equipments in welding, moulding, casting and gas cutting, such as electrodes, torches, mould boxes, patterns, furnaces, etc.
  - Perform various welding operations, such as arc welding, gas welding, spot welding, etc.
  - Perform various moulding operations, such as sand preparation, pattern making, mould making, core making, etc.
  - Perform various casting operations, such as melting, pouring, solidification, finishing, etc.
  - Perform various gas cutting operations, such as straight cutting, bevel cutting, circular cutting, etc.
  - Make simple components or assemblies by welding, moulding, casting and gas cutting, such as brackets, frames, pipes, etc.
  - Follow the safety rules and precautions while working in welding, moulding, casting and gas cutting.

- CO5 Fabricate a job by 3D printing manufacturing technique K3
  - Explain the concept and advantages of 3D printing manufacturing technique.
  - Identify and use the common types and components of 3D printers, such as FDM, SLA, SLS, extruder, nozzle, bed, etc.
  - Select and prepare the suitable materials and parameters for 3D printing, such as filament, resin, powder, temperature, speed, layer height, etc.
  - Design and model a simple component or assembly using a CAD software, such as SolidWorks, AutoCAD, etc.
  - Convert the CAD model into a 3D printable file format, such as STL, OBJ, etc.
  - Load and print the 3D printable file using a 3D printer software, such as Cura, Slic3r, etc.
  - Remove and clean the printed component or assembly from the 3D printer.
  - Follow the safety rules and precautions while working with 3D printer.

Reference Books:

- Reference books are books that contain factual information on various topics, such as dictionaries, encyclopedias, atlases, directories, handbooks, manuals, etc.
- Reference books are usually consulted for specific information, rather than read cover to cover.
- Reference books are often arranged alphabetically, chronologically, geographically, or by subject, depending on the type and purpose of the book.
- Reference books are usually kept in a separate section of a library, called the reference collection, and are not available for borrowing.
- Reference books are useful for finding quick facts, definitions, statistics, maps, illustrations, bibliographies, and other sources of information on a topic.

1. Workshop Practice, H S Bawa, McGraw Hill

- This is a book that covers the basic concepts and skills of workshop technology for engineering students.
- The book is divided into 15 chapters, each focusing on a specific topic such as fitting, sheet metal work, welding, carpentry, machine tools, etc.
- The book provides theoretical explanations, practical examples, diagrams, illustrations, and exercises to enhance the understanding and application of workshop techniques.
- The book also includes safety precautions, common workshop tools and equipment, and workshop standards and specifications.

- The book is suitable for students of mechanical, civil, electrical, and production engineering, as well as diploma and vocational courses.

2. Mechanical Workshop Practice, K C John, PHI

- This is a book that presents the basic principles of manufacturing processes and engineering materials, tools and equipment commonly used in the engineering field.
- The book covers the following topics:
  - Introduction to workshop practice
  - Safety precautions and first aid
  - Benchwork and fitting
  - Sheet metal work
  - Carpentry and pattern making
  - Foundry and casting
  - Smithy and forging
  - Welding and brazing
  - Machine tools and machining
  - Surface finishing and heat treatment
- The book is designed for the core course on Workshop Practice offered to all first-year diploma and degree level students of engineering.
- The book is written by K C John, who is a professor of mechanical engineering at the National Institute of Technology, Calicut, India.
- The book is published by PHI Learning Private Limited, which is a leading academic publisher in India.

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications

- Workshop Practice is a course that introduces the students to the basic concepts and skills of various workshop processes, such as fitting, sheet metal, soldering, welding, carpentry, etc.
- The course aims to develop the students' practical knowledge and abilities to perform different types of workshop operations and to use various tools and machines safely and efficiently.
- The course also helps the students to understand the applications and limitations of different workshop processes and materials in engineering and manufacturing.
- Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications are two books that cover the theoretical and practical aspects of workshop practice in a comprehensive and systematic manner.
- The books are divided into several chapters, each dealing with a specific workshop process, such as bench work and fitting, sheet metal work, smithy and forging, welding, carpentry and pattern making, etc.
- The books provide detailed descriptions and illustrations of the tools, equipment, materials, methods, procedures, and safety precautions in-

volved in each workshop process.
- The books also include numerous examples, exercises, and questions to test the students' understanding and to enhance their problem-solving skills.
- The books are suitable for the students of diploma, degree, and certificate courses in engineering and technology, as well as for the professionals and hobbyists who want to learn and practice workshop skills.

4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication.

- This book covers the fundamentals of computer numerical control (CNC) for various types of machines, such as lathes, milling machines, drilling machines, and machining centers.
- It also explains the principles of CNC programming, including coordinate systems, interpolation, tool compensation, canned cycles, subprograms, macros, and parametric programming.
- The book provides numerous examples and exercises to illustrate the application of CNC concepts and techniques to various machining operations and situations.
- The book also includes a chapter on computer-aided manufacturing (CAM) systems, which are software tools that automate the generation of CNC programs from CAD models or drawings.
- The book is intended for students, engineers, technicians, and operators who are involved in CNC machining or want to learn more about it.
- The book assumes a basic knowledge of engineering drawing, mathematics, and machining processes.